IDENTITY & ACCESS MANAGEMENT

IAM in Cloud Environments

Securing digital identities and protecting cloud resources through modern access controls.

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Objectives

01 Understand IAM

Define Identity and Access Management and its scope in modern cloud environments.

02 Cloud Importance

Explain why IAM is critical for securing multi-tenant cloud infrastructure.

03 Learning Goals

Explore authentication, authorization, and access control mechanisms.

04 Resource Protection

Learn best practices for safeguarding cloud data, applications, and services.

SLIDE 03

Introduction to IAM

What is IAM?

Identity and Access Management is a framework of policies and technologies that ensures the right individuals access the right resources at the right time — for the right reasons.

Why we need it

- Prevents unauthorized access to data
- Enforces least-privilege on cloud resources
- Supports compliance & audit requirements
- Enables secure remote & hybrid workforces

The IAAA Model

I

Identification

Claiming an identity (e.g. username, user ID).

A

Authentication

Proving the identity (password, MFA, biometrics).

A

Authorization

Granting permissions to specific resources.

A

Accounting

Logging and auditing user actions over time.

SLIDE 04

Core Concepts of IAM

Identity

Digital representation of a user, service, or device.

User Account

Credentials + attributes tied to a specific identity.

Authentication

Verifying an identity through credentials or factors.

Authorization

Deciding what actions an identity is allowed.

Roles

Named sets of permissions assigned to users.

Policies

Rules that define allowed or denied actions.

Permissions

Fine-grained rights over specific resources.

Least Privilege

Grant only the minimum access required.

RBAC

Role-Based Access Control — access via job role.

ABAC

Attribute-Based Access Control — decisions from user/resource attrs.

MFA

Multi-Factor Authentication — 2+ verification factors.

SSO & Federation

One login across trusted systems and clouds.

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IAM Architecture

1. **User**
   End user requesting access
2. **Login Portal**
   Entry point / web UI
3. **Identity Provider (IdP)**
   Manages identities & credentials
4. **Authentication**
   Password + MFA verification
5. **IAM Service**
   Central policy decision point
6. **Authorization Policies**
   Evaluates allow / deny rules

Cloud Resources

Access granted after policy evaluation

Virtual Machines

Compute instances

Databases

Managed SQL / NoSQL

Storage

Object & block storage

Applications

SaaS & internal apps

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How IAM Works — Step by Step

STEP 1

User Requests Access

The user attempts to access a cloud resource or service.

STEP 2

Identity Verification

IAM identifies the account attempting the request.

STEP 3

Password + MFA

Credentials plus a second factor are verified.

STEP 4

Policy Evaluation

IAM engine checks roles, permissions and conditions.

STEP 5

Permit or Deny

Decision returned based on applicable policies.

STEP 6

Access Resource

On permit, the user interacts with the cloud service.

SLIDE 07

Authentication Methods

P

Password

Traditional secret-based login. Simple but vulnerable to reuse and phishing.

M

Multi-Factor (MFA)

Combines two or more factors: something you know, have, or are.

B

Biometric

Fingerprint, face or voice recognition — tied to physical traits.

O

One-Time Password (OTP)

Time-based codes via authenticator apps or SMS.

K

Security Keys

Hardware tokens (FIDO2 / WebAuthn) resistant to phishing.

S

Single Sign-On (SSO)

One authenticated session across many trusted apps.

F

Identity Federation

Trust between orgs / clouds using SAML, OIDC, OAuth 2.0.

*

Passwordless

Magic links, passkeys, device biometrics — no password stored.

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Authorization Models

Model	How it works	Best for
RBAC	Access assigned via roles mapped to job functions.	Stable teams with clear job roles.
ABAC	Decisions from attributes: user, resource, context.	Dynamic, fine-grained access at scale.
Policy-Based	Explicit JSON / rule policies attached to identities.	Central governance & auditing.
Least Privilege	Grant only the minimum rights required for a task.	Reducing blast radius of compromise.
Zero Trust	Never trust, always verify — every request re-checked.	Hybrid / remote / multi-cloud.

In practice: Cloud providers combine RBAC + ABAC + policy documents under a Zero Trust mindset, enforcing least privilege by default.

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Security Benefits of IAM

✓

Improved Security

Strong authentication and centralized control reduce breach risk.

✓

Reduced Insider Threats

Least privilege limits what compromised accounts can do.

✓

Compliance Support

Meets GDPR, HIPAA, ISO 27001, SOC 2 audit requirements.

✓

Centralized Management

Single pane for users, roles, groups and policies.

✓

Secure Remote Access

Enables hybrid work without exposing the perimeter.

✓

Audit Logs

Every access decision is recorded for forensic review.

✓

Risk Reduction

Adaptive policies react to unusual behavior in real time.

✓

Data Protection

Fine-grained control safeguards sensitive information.

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Enterprise IAM Implementations

AWS

AWS IAM

Users, groups, roles and JSON policies with fine-grained resource control. Supports IAM Identity Center (SSO) and STS temporary credentials.

AZ

Microsoft Entra ID

Azure's identity platform (formerly Azure AD). Conditional Access, MFA, RBAC on Azure resources and B2B/B2C federation.

GC

Google Cloud IAM

Principal + Role + Resource model with hierarchical policies across Organization → Folder → Project → Resource.

IBM

IBM Cloud IAM

Unified access via access groups, service IDs and trusted profiles across IBM Cloud services with policy-based control.

Practical Example — University Cloud Portal

Student

S

View assignments, submit work, check grades.

Faculty

F

Upload marks, manage courses, view rosters.

Administrator

A

Manage users, roles, and system policies.

University IAM

Authenticate + Authorize

SSO • MFA • RBAC

Assignments

Student read/write

Grades DB

Faculty write, Student read

User Directory

Admin only

Course Content

Faculty write, all read

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Challenges, Future Scope & Conclusion

CHALLENGES

Limitations of IAM

- Complex policy management
- Misconfigured permissions
- Vendor lock-in / dependency
- Identity-based attacks (phishing, token theft)
- Scaling roles across teams

FUTURE SCOPE

Where IAM is heading

- AI-powered identity management
- Passwordless authentication
- Behavior-based authentication
- Zero Trust Architecture
- Decentralized identity (blockchain)
- Adaptive & continuous authentication

CONCLUSION

Key Takeaways

- IAM = right access, right user, right time.
- Essential for securing cloud resources.
- Enables compliance & remote work.
- Adopted by AWS, Azure, GCP, IBM Cloud.
- Future = passwordless + Zero Trust + AI.

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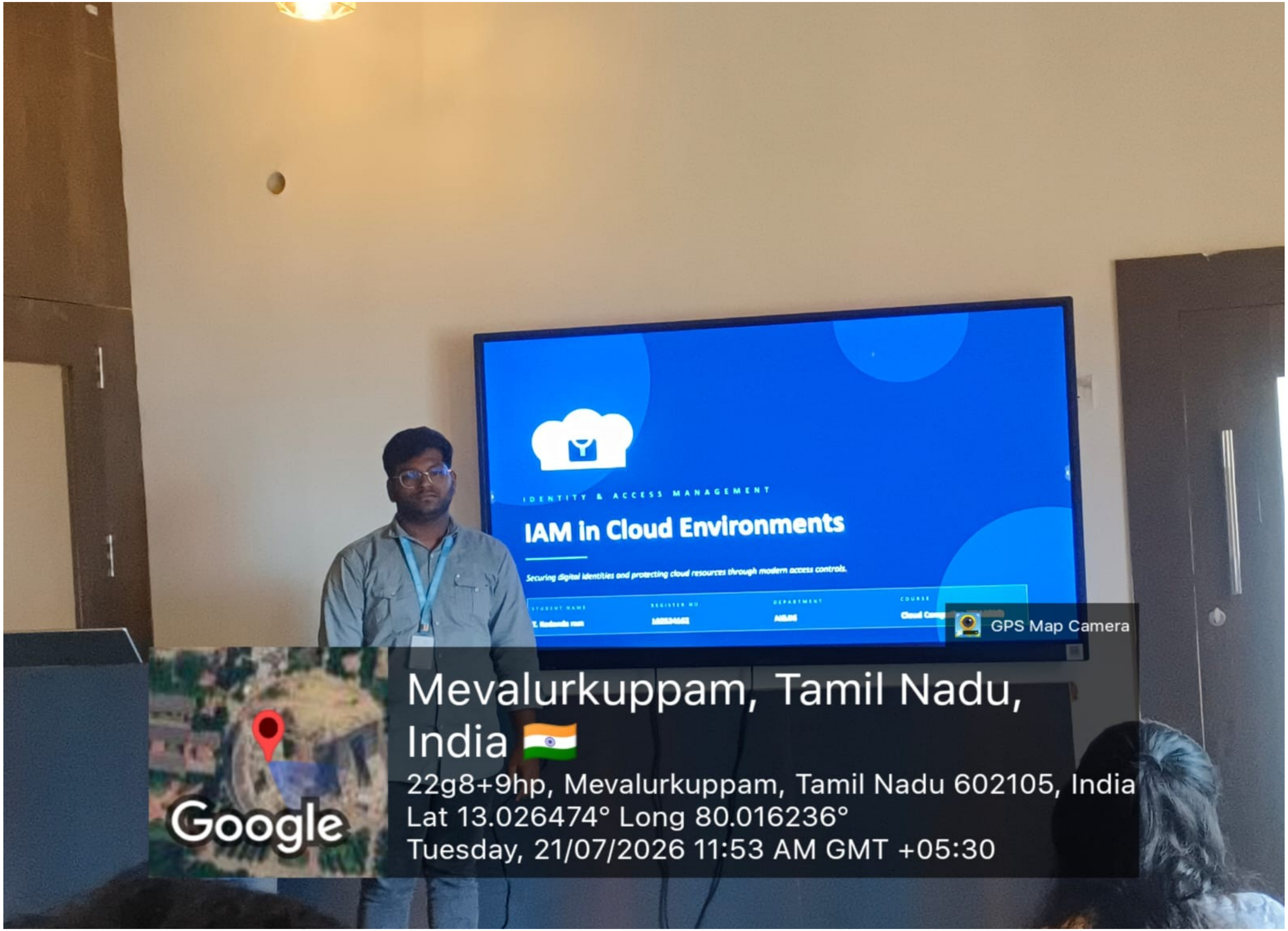
Thank you — Questions?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